A CORRESPONDENCE FOR THE GENERALIZED HECKE ALGEBRA OF THE METAPLECTIC COVER $\overline{SL(2,F)}$, F p-ADIC

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ABSTRACT. We prove, using a technique developed for $GL(n)$ in Howe and Moy [H], a bijection between generalized Hecke algebras of $G = SL(2,F)$ over a p-adic field and those of its n-fold metaplectic cover $\overline{G}$. This result implies that there is a canonical correspondence between irreducible admissible representations of G and genuine irreducible admissible representations of $\overline{G}$ of "sufficiently large level" (depending on n, p).

CONTENTS

1. Some notation 2
2. Some lemmas 3
3. The structure of $\mathcal{H}(G//B_k)$ 6
4. The structure of $\mathcal{H}(\overline{G}//B_k)_{gen}$ 9
5. Application to representation theory 10
6. The Iwahori algebra 13
References 13

Let F be a p-adic field of characteristic 0, with uniformizer π, ring of integers $\mathcal{O}$, and let q denote the cardinality of the residue field. Let $\mu_n(F)$ denote the group of nth roots of unity in F and assume $|\mu_n(F)| = n$. We will identify $\mu_n(F)$ with a subgroup of $\mathbb{C}^\times$ (via some fixed isomorphism $\theta : \mu_n(F) \to \mu_n(\mathbb{C})$). Write $\mu_n = \mu_n(F)$. Let $G = SL(2,F)$ and let $\mathfrak{g}$ denote its Lie algebra. Let $\overline{G} = \overline{SL(2,F)}$ denote the n-fold metaplectic cover defined by the Kubota cocycle

$$\beta(g_1, g_2) = \left(\frac{x(g_1 g_2)}{x(g_1)}, \frac{x(g_1 g_2)}{x(g_2)} \right),$$

for $g_1, g_2 \in G$, where

$$x\begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{cases} c, & \text{if } c \neq 0, \\ d, & \text{otherwise,} \end{cases}$$

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and where $(\ ,\ )$ is the nth power Hilbert symbol on $F^\times \times F^\times$. By §1 of Kubota [K], β is a 2-cocycle (called a "factor set" in [K]) which is trivial near the identity in $G \times G$, so $\overline{G}$ is a topological covering group of G. We denote elements of $\overline{G}$ by $\overline{x} = (x, \zeta)$, $x \in G$, $\zeta \in \mu_n$, and multiplication by

$$\overline{x}_1 \overline{x}_2 = (x_1, \zeta_1)(x_2, \zeta_2) = (x_1 x_2, \beta(x_1, x_2)\zeta_1 \zeta_2).$$

Let $\rho : \overline{G} \to G$ denote the natural surjection. If H is any subgroup of G then we define $\overline{H} = \rho^{-1}(H)$. We choose a Haar measure on G, to be normalized below, and a Haar measure on μ_n normalized so that $\mathrm{vol}(\mu_n) = 1$. Let the Haar measure on $\overline{G}$ be the product measure.

We prove, using a technique developed for $GL(n)$ in [H], a bijection between generalized Hecke algebras of G and those of $\overline{G}$. A precise statement will be given in Theorem 11. This result implies that there is a canonical one-to-one correspondence between irreducible admissible representations of G and genuine irreducible admissible representations of $\overline{G}$ of "sufficiently large level" (depending on n, p). For the precise statement, see Corollary 12.

The study of genuine admissible representations on p-adic groups has its origins in the book [G] by S. Gelbart. Flicker [F] determined a comparison of Hecke algebras of *smooth functions* on $GL(2, F)$ and $\overline{GL(2, F)}$ using the matching of orbital integrals. This was then used, with the Arthur-Selberg trace formula, to establish a global "Shimura correspondence". Neither of these two works used the Hecke algebras considered here. G. Savin [S] has given an isomorphism between the Iwahori-Hecke algebras in the case when p is relatively prime to $2n$ and G is any reductive p-adic group which splits over $\mathcal{O}$. This isomorphism generalized a result of Flicker and Kazhdan [FK] who considered the special case of $GL(r)$. (Savin's result is discussed briefly in §6 below.) The works of Rallis-Schiffmann [RS] and Schultz [Sch] also dealt with a correspondence between representations on $\overline{G}$ and on G (or a closely related group), though from a different perspective and using different methods. Our hope is that the result in the present paper will form one small step, along with Waldspurger's multiplicity one theorem for $\overline{SL(2, F)}$ [W] and the Arthur-Selberg trace formula, in an eventual proof of Labesse's multiplicity one conjecture for $SL(2)$.

1. Some notation

Let

$$s_1 = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad s_2 = \begin{pmatrix} 0 & -\pi^{-1} \\ \pi & 0 \end{pmatrix}, \quad t = \begin{pmatrix} 0 & -1 \\ \pi & 0 \end{pmatrix}, \quad t_0 = \begin{pmatrix} \pi & 0 \\ 0 & \pi^{-1} \end{pmatrix}.$$

Note $t_0 = s_1^{-1} s_2$, $t \notin G$, and $t s_1 t^{-1} = s_2$. Although $t \notin G$, conjugation by t defines an automorphism of G of order 2 (since $t^2 = -\pi I$ is central in $GL(2, F)$).

Let

$$u_+(x) = \begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix}, \quad u_-(x) = \begin{pmatrix} 1 & 0 \\ x & 1 \end{pmatrix}, \quad h(x) = \begin{pmatrix} x & 0 \\ 0 & x^{-1} \end{pmatrix}.$$

Let

$$B = \left\{ \begin{pmatrix} \mathcal{O} & \mathcal{O} \\ \pi\mathcal{O} & \mathcal{O} \end{pmatrix} \right\} \cap G$$

denote the Iwahori subgroup,

$$\mathfrak{b} = \left\{ \begin{pmatrix} \mathcal{O} & \mathcal{O} \\ \pi\mathcal{O} & \mathcal{O} \end{pmatrix} \right\} \cap \mathfrak{g}$$

denote its Lie algebra, and for $k > 0$ let

$$B_k = \left(I + t^{2k}\begin{pmatrix}\mathcal{O} & \mathcal{O}\\ \pi\mathcal{O} & \mathcal{O}\end{pmatrix}\right) \cap G = \left\{\begin{pmatrix}a & b\\ \pi c & d\end{pmatrix} \in G \;\middle|\; a-1,\, d-1,\, b,\, c \in \pi^k\mathcal{O}\right\}$$

denote the kth filtration subgroup of B. Note that $tBt^{-1} = B$ and $tB_kt^{-1} = B_k$. Let

$$\mathfrak{b}_k = \left\{\begin{pmatrix}a & b\\ \pi c & -a\end{pmatrix} \;\middle|\; a, b, c \in \pi^k\mathcal{O}\right\}$$

denote the Lie algebra of B_k.

Definition 1. Let H denote a locally compact unimodular group and let K denote an open compact subgroup of H with Haar measure on H normalized so that K has measure one. Let $\mathcal{H}(H//K)$ denote the space of functions on H which are locally constant, compactly supported, and bi-K-invariant. This forms an algebra under convolution,

$$(f * g)(x) = \int_H f(y)g(y^{-1}x)\,dy,$$

called the **generalized Hecke algebra**.

We shall study the algebra $\mathcal{H}(G//B_k)$ in Section 3.

Recall that a 2-cocycle $\alpha : H \times H \to \mu_n$ of a group H is **trivial** if there is a function $s : H \to \mu_n$ (called a **splitting**) such that $\alpha(h_1, h_2) = s(h_1)s(h_2)s(h_1h_2)^{-1}$, for $h_1, h_2 \in H$. In this case the cover of H defined by α is said to **split**, and the map $h \mapsto (h, s(h)^{-1})$ embeds H as a subgroup of $\overline{H}$. When α is replaced by the equivalent cocycle $\alpha'(h_1, h_2) = \alpha(h_1, h_2)s(h_1)^{-1}s(h_2)^{-1}s(h_1h_2)$ on $H \times H$, the splitting becomes trivial on H, and H embeds canonically as $H^* = \{(h, 1) \mid h \in H\} \subset \overline{H}$.

By Hensel's Lemma, if $k > 2\,\mathrm{ord}_\pi(n)$ (and in particular for any $k \geq 1$ when $\gcd(p, n) = 1$), we have $1 + \pi^k\mathcal{O} \subset (F^\times)^n$. For such k, every $u \in 1 + \pi^k\mathcal{O}$ is an nth power in $F^\times$, so the nth power Hilbert symbol satisfies $(u, y) = 1$ for all $u \in 1 + \pi^k\mathcal{O}$ and all $y \in F^\times$. By [K], Theorem 2, the Kubota splitting function $s_K : G \to \mu_n$ defined by

$$s_K\begin{pmatrix}a & b\\ c & d\end{pmatrix} = \begin{cases}(c, d), & \text{if } cd \neq 0 \text{ and } \mathrm{ord}_\pi(c) \text{ is odd},\\ 1, & \text{otherwise},\end{cases}$$

trivializes β on the principal congruence subgroup of level π^k, and satisfies $s_K|_{B_k} \equiv 1$ since $d \in 1 + \pi^k\mathcal{O} \subset (F^\times)^n$ for all elements of B_k. In fact, directly from the definition of β, for any $g_1, g_2 \in B_k$, the Iwahori factorization $B_k = U_k^- T_k U_k^+$ implies that $\overline{B}_k$ splits.

Recall that a complex-valued function f on $\overline{G}$ is called **genuine** if $f(g, \zeta) = \theta(\zeta)f(g, 1)$, for $g \in G$ and $\zeta \in \mu_n$. Here θ is a fixed isomorphism $\theta : \mu_n \to \mu_n(\mathbb{C})$. It being agreed that we may identify μ_n and $\mu_n(\mathbb{C})$, we shall, *from this point on, drop θ from the notation.*

Definition 2. Assume that $k > 0$ is chosen so large that $1 + \pi^k\mathcal{O} \subset (F^\times)^n$ (so in particular $\overline{B}_k$ splits). Fix a cocycle β' equivalent to β such that $\beta'|_{B_k \times B_k} = 1$, and identify B_k with the split subgroup $B_k^* = \{(b, 1) \mid b \in B_k\} \subset \overline{B}_k$. Assume that the Haar measure on $\overline{G}$ is normalized so that $\overline{B}_k$ has measure 1. Let $\mathcal{H}(\overline{G}//B_k)_{gen}$ denote the space of functions on $\overline{G}$ which are genuine, locally constant, compactly supported, and bi-B_k^*-invariant. This is the **(metaplectic) generalized Hecke algebra** on $\overline{G}$.

We will study this algebra in Section 4. Note that it depends on the splitting of the cocycle on B_k. To have a measure on $\overline{G}$ which fits with the measure on G used in §3, we shall also assume that μ_n has measure 1.

2. Some lemmas

Let $\mathbb{F}_q = \mathcal{O}/\pi\mathcal{O}$ denote the residue field of F.

Lemma 3. *For $k > 0$, $B_k/B_{k+1} \cong \mathfrak{b}_k/\mathfrak{b}_{k+1} \cong \mathbb{F}_q^3$, as abelian groups. (The second isomorphism is not canonical.)*

Proof. The graph of the first isomorphism is $\{(g\cdot B_{k+1}, X+\mathfrak{b}_{k+1}) \mid (I+X+\mathfrak{b}_{k+1})\cap g\cdot B_{k+1} \neq \emptyset\}$ (namely, $I + X \bmod B_{k+1} \mapsto X \bmod \mathfrak{b}_{k+1}$ for $X \in \mathfrak{b}_k$). It is also a special case of a general result of Morris (Proposition 3.4 in [M]). The second isomorphism is due to the fact that the surjective homomorphism

$$\begin{pmatrix} a & b \\ \pi c & -a \end{pmatrix} \longmapsto \left(\pi^{-k}a \bmod \pi,\ \pi^{-k}b \bmod \pi,\ \pi^{-k}c \bmod \pi\right) \in \mathbb{F}_q^3$$

has image $\mathbb{F}_q^3$ and kernel $\mathfrak{b}_{k+1}$. □

The following result is stated as Proposition 2.2 in Chapter 3 of [H].

Lemma 4. *In the notation of Definition 1, let f_h denote the characteristic function of the double coset KhK in H. Then f_1 is the unit of $\mathcal{H}(H//K)$. More generally, if*

$$\mathrm{vol}(Kh_1K)\,\mathrm{vol}(Kh_2K) = \mathrm{vol}(Kh_1h_2K),$$

then $f_{h_1} * f_{h_2} = f_{h_1h_2}$.

We need a version of the previous lemma for genuine functions. Assume β' is chosen as in Definition 2 so that $\beta'|_{B_k\times B_k} = 1$. Since $1+\pi^k\mathcal{O} \subset (F^\times)^n$, for any $g \in G$ and any $b_1, b_2 \in B_k$ satisfying $b_1gb_2 = g$, the compatibility of the Kubota splitting with $1 + \pi^k\mathcal{O} \subset (F^\times)^n$ ensures that $(b_1, 1)(g, 1)(b_2, 1) = (g, 1)$. Thus, for each $g \in G$, there is a unique genuine bi-B_k^*-invariant function $\overline{f}_g$ supported on $\overline{B}_k(g,1)\overline{B}_k$ satisfying $\overline{f}_g(g, 1) = 1$; explicitly, for $b_1, b_2 \in B_k$ and $\zeta \in \mu_n$,

$$\overline{f}_g\big((b_1, 1)(g, 1)(b_2, 1)(1, \zeta)\big) = \zeta,$$

and $\overline{f}_g(\overline{x}) = 0$ for $\overline{x} \notin \overline{B}_k(g,1)\overline{B}_k$. When β' is additionally normalized on the double coset B_kgB_k so that $\beta'(b_1, g)\beta'(b_1g, b_2) = 1$ for $b_1, b_2 \in B_k$, this takes the simple form

$$\overline{f}_g(x, \zeta) = \begin{cases} \zeta, & x \in B_kgB_k, \\ 0, & \text{otherwise.} \end{cases}$$

Let

$$(\phi_1 * \phi_2)(g) = \int_{\overline{G}} \phi_1(x)\phi_2(x^{-1}g)\,dx, \quad g \in \overline{G}.$$

Now we prove a genuine analog of Lemma 4 above.

Lemma 5. *Assume $\mathrm{vol}(\overline{B}_k) = 1$. If $g_1, g_2 \in G$ are such that $\beta'((g_1, 1), (g_2, 1)) = 1$ on the double coset representatives and*

$$\mathrm{vol}(B_kg_1B_k)\,\mathrm{vol}(B_kg_2B_k) = \mathrm{vol}(B_kg_1g_2B_k),$$

then $\overline{f}_{g_1} * \overline{f}_{g_2} = \overline{f}_{g_1g_2}$.

Proof. We compute

$$(\overline{f}_{g_1} * \overline{f}_{g_2})(g,1) = \int_{\overline{G}} \overline{f}_{g_1}(x)\overline{f}_{g_2}(x^{-1}(g,1))\,dx$$
$$= \int_G \overline{f}_{g_1}(x,1)\overline{f}_{g_2}(x^{-1}g,1)\beta'(x,x^{-1}g)^{-1}\,dx.$$

Since $|\overline{f}_{g_i}(x,1)| = f_{g_i}(x)$, the support of this integral is contained in the support of $f_{g_1} * f_{g_2}$. By Lemma 4, $f_{g_1} * f_{g_2} = f_{g_1g_2}$, so

$$\mathrm{supp}(\overline{f}_{g_1} * \overline{f}_{g_2}) \subset \mathrm{supp}(\overline{f}_{g_1g_2}).$$

Since $\overline{f}_{g_1} * \overline{f}_{g_2}$ is genuine and bi-B_k^*-invariant, and the space of genuine bi-B_k^*-invariant functions supported on a single double coset $\overline{B}_k(g_1g_2,1)\overline{B}_k$ is one-dimensional (spanned by $\overline{f}_{g_1g_2}$), we must have $\overline{f}_{g_1} * \overline{f}_{g_2} = c\overline{f}_{g_1g_2}$ for some constant c. Evaluating at $(g_1g_2,1)$: by the volume condition $\mathrm{vol}(B_kg_1B_k)\,\mathrm{vol}(B_kg_2B_k) = \mathrm{vol}(B_kg_1g_2B_k)$, the map $B_k \times B_k \times B_k \to B_kg_1g_2B_k$ given by $(b_1,b_2,b_3) \mapsto b_1g_1b_2g_2b_3$ has fiber B_k at g_1g_2, so for $x = g_1b_2$ ($b_2 \in B_k$) we have $x^{-1}g_1g_2 = b_2^{-1}g_2$ and the integrand at $g = g_1g_2$ is identically 1 on g_1B_k, yielding $c = \mathrm{vol}(g_1B_k) = 1$. □

As in the above proof, we have the following expression for the convolution in terms of the cocycle.

Lemma 6. *For any genuine functions $\phi_1, \phi_2 \in \mathcal{H}(\overline{G}//B_k)_{gen}$, we have*

$$(\phi_1 * \phi_2)(g,\zeta) = \zeta \int_G \phi_1(x,1)\phi_2(x^{-1}g,1)\beta'(x,x^{-1}g)^{-1}\,dx,$$

for any $g \in G$ and $\zeta \in \mu_n$.

Note that the generalized metaplectic Hecke algebra depends on the cocycle β, so let us denote this dependence temporarily by $\mathcal{H}_\beta(\overline{G}//B_k)_{gen}$. If the cocycle is changed to an equivalent one, say to β', where

$$\beta(g,h) = \beta'(g,h)s(g)s(h)s(gh)^{-1}$$

is the cocycle modified by s and $s : G \to \mu_n$ is any function satisfying $s(1) = 1$ and $s|_{B_k} \equiv 1$ (for example, the Kubota splitting [K, KP]), then the two algebras are isomorphic.

Lemma 7. *Let β, β' be as above and let $s : G \to \mu_n$ satisfy $s(1) = 1$. There is a canonical isomorphism*

$$\phi : \mathcal{H}_{\beta'}(\overline{G}//B_k)_{gen} \xrightarrow{\sim} \mathcal{H}_\beta(\overline{G}//B_k)_{gen},$$

as algebras, defined by sending $f(g,\zeta) = \zeta f(g,1)$ to $\phi(f)(g,\zeta) = \zeta s(g)f(g,1)$. In other words, under this mapping, we have

$$\phi(f_1 *_{\beta'} f_2)(g,\zeta) = (\phi(f_1) *_\beta \phi(f_2))(g,\zeta),$$

where $*_\beta$ denotes the convolution with respect to the β cocycle

$$(\phi_1 *_\beta \phi_2)(g,\zeta) = \zeta \int_G \phi_1(x,1)\phi_2(x^{-1}g,1)\beta(x,x^{-1}g)^{-1}\,dx$$

and, similarly, $*_{\beta'}$ denotes the convolution with respect to the β' cocycle.

Proof. Using $\beta(x,x^{-1}g)^{-1} = \beta'(x,x^{-1}g)^{-1}s(x)^{-1}s(x^{-1}g)^{-1}s(g)$, we have

$$\begin{aligned}(\phi(f_1)*_\beta \phi(f_2))(g,\zeta) &= \zeta\int_G s(x)f_1(x,1)s(x^{-1}g)f_2(x^{-1}g,1)\beta(x,x^{-1}g)^{-1}\,dx \\ &= \zeta s(g)\int_G f_1(x,1)f_2(x^{-1}g,1)\beta'(x,x^{-1}g)^{-1}\,dx \\ &= \phi(f_1 *_{\beta'} f_2)(g,\zeta),\end{aligned}$$

as claimed. □

3. The structure of $\mathcal{H}(G//B_k)$

We want to determine the structure of the generalized Hecke algebra as a finitely generated algebra with generators and relations. Then we will do the same for the metaplectic analog and compare the two.

The affine Weyl group $W_a = N_G(T)/(T\cap B)$ of G is an infinite dihedral group generated by the images of s_1 and s_2, and the finite Weyl group W is generated by the image of s_1. Note that in G, $s_1^2 = s_2^2 = -I \in B$, and every element of the group $\langle s_1, s_2\rangle \subset G$ can be written up to sign $\pm I \in B$ as a reduced alternating word in s_1 and s_2 (in particular, $t_0 = s_1^{-1}s_2 = h(-1)s_1s_2$). We will use the Iwahori-Bruhat decomposition

$$G = BW_aB$$

to determine the structure of the generalized Hecke algebra.

Recall $(\operatorname{Ad} g)(x) = gxg^{-1}$. Let $S = \{s_1, s_2\}\cup B$.

Proposition 8. *The algebra $\mathcal{H}(G//B_k)$ is generated by the functions f_g, $g\in S$. Assume that $\operatorname{vol}(B_k) = 1$. These elements are subject to the relations below.*

(A) Relations for the s_i's ($i = 1,2$):

(i) *$f_{s_1} * f_{s_1} = c_1\sum_x f_x$, where x runs over a set of representatives of $((-1)\cdot(\operatorname{Ad} s_1)(B_k)\cdot B_k)/B_k$ and $c_1 = \operatorname{vol}(B_k s_1 B_k) = q$,*

(ii) *$f_{s_2} * f_{s_2} = c_2\sum_x f_x$, where x runs over a set of representatives of $((-1)\cdot(\operatorname{Ad} s_2)(B_k)\cdot B_k)/B_k$ and $c_2 = \operatorname{vol}(B_k s_2 B_k) = q$.*

(B) Relations for elements of B:

(i) *f_1 is the identity of $\mathcal{H}(G//B_k)$,*

(ii) *$f_x * f_y = f_{xy}$, for $x,y\in B$,*

(iii) *$f_{xg} = f_{gx} = f_g$, for all $x\in B_k$, $g\in G$.*

(C) Mixed relations:

(i) *$f_{s_i} * f_x = f_{\operatorname{Ad} s_i(x)} * f_{s_i}$, for $x\in B\cap(\operatorname{Ad} s_i^{-1})(B)$ ($i = 1,2$),*

(ii) *$f_{s_1} * f_{u_+(x)} * f_{s_1} = c_1^2 f_{\operatorname{Ad} s_1(u_+(x))} = c_1^2 f_{u_-(-x)}$, for $x\in\mathcal{O}^\times$, where $c_1 = \operatorname{vol}(B_k s_1 B_k)$,*

(iii) *$f_{s_2} * f_{u_-(-\pi x)} * f_{s_2} = c_2^2 f_{\operatorname{Ad} s_2(u_-(-\pi x))} = c_2^2 f_{u_+(x)}$, for $x\in\mathcal{O}^\times$, where $c_2 = \operatorname{vol}(B_k s_2 B_k)$.*

(The relations C(ii) and C(iii) must be expressed in terms of the generators S, as in [H]. Using the Bruhat identity $u_-(-x) = u_+(-x^{-1})s_1h(-x)u_+(-x^{-1})$ for $x\in\mathcal{O}^\times$, the relation corresponding to C(ii) is

$$f_{s_1} * f_{u_+(x)} * f_{s_1} = c_1^2\big(f_{u_+(-x^{-1})} * f_{s_1} * f_{h(-x)} * f_{u_+(-x^{-1})}\big).$$

Conjugating by t, where $ts_1t^{-1} = s_2$, $tu_+(y)t^{-1} = u_-(-\pi y)$, and $th(z)t^{-1} = h(z^{-1})$, the relation corresponding to C(iii) is

$$f_{s_2} * f_{u_-(-\pi x)} * f_{s_2} = c_2^2\big(f_{u_-(\pi x^{-1})} * f_{s_2} * f_{h(-x^{-1})} * f_{u_-(\pi x^{-1})}\big),$$

and similarly $f_{s_2} * f_{u_+(x)} * f_{s_2} = c_2^2 f_{u_-(\pi^2 x)}$ *for* $x \in \mathcal{O}^\times$*.)*

Furthermore, these relations form a defining set of relations for $\mathcal{H}(G//B_k)$ *as an algebra.*

Remark 1. (1) This proposition has an analog for G replaced by $GL(n)$, proved in [H], Theorem 2.1, Chapter 3. Unlike the $GL(n)$ case where the length-zero element t lies in $GL(n,F)$ and normalizes B_k, in $SL(2,F)$ the element $t_0 = s_1^{-1}s_2 = \mathrm{diag}(\pi, \pi^{-1})$ has length 2 in W_a and does not normalize B_k. However, the outer automorphism $\mathrm{Ad}(t)$ of $SL(2,F)$ preserves B and B_k and swaps s_1 and s_2, allowing us to use the Coxeter generators $\{s_1, s_2\}$ directly. Note that for $t_0 = h(-1)s_1s_2$, length additivity gives $f_{t_0} = f_{h(-1)} * f_{s_1} * f_{s_2}$ and $f_{t_0^{-1}} = f_{h(-1)} * f_{s_2} * f_{s_1}$, which satisfy $f_{t_0} * f_{s_1} = f_{s_1} * f_{t_0^{-1}}$.

(2) Note that though S is infinite, the set of generators $\{f_g \mid g \in S\}$ is finite (since f_b for $b \in B$ depends only on the coset $bB_k \in B/B_k$), so the algebra is indeed finitely generated.

(3) Possibly B(iii) is a superfluous relation once restricted to B/B_k.

(4) Note that s_1 and s_2 have order 4 in G ($s_1^2 = s_2^2 = -I \in B$), not order 2 as the analogous permutation matrices in [H] do.

(5) The volume constants are computed as follows:

$$\mathrm{vol}(B_k s_1 B_k) = \mathrm{vol}(B_k s_1 B_k s_1^{-1}) / \mathrm{vol}(B_k) = |U_k^- B_k / B_k| = |U_k^- / (U_k^- \cap B_k)| = |\pi^k \mathcal{O} / \pi^{k+1}\mathcal{O}| = q,$$

and since $tB_kt^{-1} = B_k$ and $ts_1t^{-1} = s_2$,

$$\mathrm{vol}(B_k s_2 B_k) = \mathrm{vol}(tB_k s_1 B_k t^{-1}) = \mathrm{vol}(B_k s_1 B_k) = q.$$

More generally, for any reduced alternating word $w = s_{i_1}s_{i_2}\cdots s_{i_r}$ ($i_j \in \{1,2\}$, $i_j \neq i_{j+1}$) and any $b \in B$, Iwahori factorization for B_k gives

$$\mathrm{vol}(B_k b w B_k) = \mathrm{vol}(B_k w B_k) = \prod_{j=1}^{r} \mathrm{vol}(B_k s_{i_j} B_k) = q^r,$$

so by Lemma 4, $f_w = f_{s_{i_1}} * f_{s_{i_2}} * \cdots * f_{s_{i_r}}$.

Here are some general facts we will use in the proof below.

Lemma 9. (a) *B_k is a normal subgroup of B, and $tB_kt^{-1} = B_k$.*

(b) *Let $U_k^+ = \{u_+(x) \mid x \in \pi^k\mathcal{O}\}$ and $U_k^- = \{u_-(x) \mid x \in \pi^k\mathcal{O}\}$. We have*

$$B_k \cdot (s_1^{-1}B_k s_1) = (s_1^{-1}B_k s_1) \cdot B_k = U_k^- B_k,$$

and

$$B_k \cdot (s_2^{-1}B_k s_2) = (s_2^{-1}B_k s_2) \cdot B_k = U_{k-1}^+ B_k,$$

and both $U_k^- B_k$ and $U_{k-1}^+ B_k$ are groups containing B_k as a normal subgroup of index q.

(c) *In general, we have*

$$(f_{g_1} * f_{g_2})(x) = \int_G f_{g_1}(y) f_{g_2}(y^{-1}x)\, dy = \mathrm{vol}(B_k g_1 B_k \cap x B_k g_2^{-1} B_k).$$

(d) *In general, we have*

$$\int_G (f*g)(x)\, dx = \int_G f(y)\, dy \int_G g(x)\, dx, \quad \int_G (f*g*h)(x)\, dx = \int_G f(x)\, dx \int_G g(y)\, dy \int_G h(z)\, dz.$$

Proof. Parts (a) and (b) may be proven by direct matrix calculations (with the second identity in (b) following from the first by conjugating by t). Parts (c) and (d) are simple consequences of the definitions. □

Proof of Proposition 8. We shall follow the ideas in [H], proof of Theorem 2.1.

We begin by verifying A(i): $f_{s_1} * f_{s_1} = c_1 \sum_{x \in X} f_x$, where x runs over a complete set X of representatives of $((-1) \cdot (\mathrm{Ad}\, s_1)(B_k) \cdot B_k)/B_k$. As in [H], parts (a) and (b) of Lemma 9 imply that

$$f_{s_1} * f_{s_1} = c_1 \cdot \chi_{(-1) \cdot U_k^- B_k},$$

where χ_E denotes the characteristic function of a subset E of G and where $c_1 \neq 0$ is some constant. In other words,

$$(f_{s_1} * f_{s_1})(g) = c_1 \sum_{x \in X} f_x(g), \quad g \in G.$$

Plugging $g = -I = s_1^2$ into both sides and using Lemma 9(c), we obtain

$$\mathrm{vol}(B_k s_1 B_k) = c_1.$$

Conjugating by the measure-preserving automorphism $\mathrm{Ad}(t)$ (which maps B_k to B_k and s_1 to s_2) immediately gives A(ii) with $c_2 = \mathrm{vol}(B_k s_2 B_k) = q$.

Part (c) of Lemma 9 implies part B(i) of the proposition.

We now verify part B(ii). We know $\mathrm{vol}(B_k) = 1$ implies $\mathrm{vol}(bB_k) = 1$, for each $b \in B$. Since B_k is normal in B, $B_k b B_k = bB_k$, so Lemma 4 implies part B(ii).

Part B(iii) follows from the definition of f_g.

We shall now verify C(i). Since B_k is normal in B, for any $b \in B$ we have

$$\mathrm{vol}(B_k s_i b B_k) = \mathrm{vol}(B_k s_i b B_k b^{-1} b) = \mathrm{vol}(B_k s_i B_k b) = \mathrm{vol}(B_k s_i B_k).$$

This and Lemma 4 imply $f_{s_i} * f_x = f_{s_i x}$, for $x \in B$. Likewise, if $s_i x s_i^{-1} \in B$ then $f_{s_i x s_i^{-1}} * f_{s_i} = f_{s_i x}$. This implies C(i).

We next verify C(ii). The proof of this part is sketched in [H]. The argument in [H] gives $f_{s_1} * f_{u_+(x)} * f_{s_1} = c f_{u_-(-x)}$, for $x \in \mathcal{O}^\times$, for some constant $c \neq 0$. Here c may be evaluated using part (d) of Lemma 9, which implies

$$c = \frac{\mathrm{vol}(B_k s_1 B_k)^2 \, \mathrm{vol}(B_k u_+(x) B_k)}{\mathrm{vol}(B_k u_-(-x) B_k)} = \mathrm{vol}(B_k s_1 B_k)^2 = c_1^2.$$

(Here we have used the fact that $u_+(x) \in B$, so $\mathrm{vol}(B_k u_+(x) B_k) = \mathrm{vol}(B_k u_+(x)) = \mathrm{vol}(B_k) = 1$, and the fact that $\mathrm{vol}(B_k u_-(-x) B_k) = \mathrm{vol}(s_1^{-1} B_k s_1 u_+(x) s_1^{-1} B_k s_1) = \mathrm{vol}(s_1^{-1} B_k s_1 u_+(x)) = \mathrm{vol}(s_1^{-1} B_k s_1) = 1$.) Since $u_-(-x) = u_+(-x^{-1}) s_1 h(-x) u_+(-x^{-1})$ with $u_+(-x^{-1}), h(-x) \in B$, we have $f_{u_-(-x)} = f_{u_+(-x^{-1})} * f_{s_1} * f_{h(-x)} * f_{u_+(-x^{-1})}$ by C(i) and B(ii).

Part C(iii) follows directly from C(ii) by applying the automorphism $\mathrm{Ad}(t)$.

It remains to show that this list of relations is a defining set of relations for the algebra. To this end, we follow the proof of [H], Chapter 3, Theorem 2.1, as follows.

For $g \in S$, let $\tilde{f}_g$ denote an abstract element. Let $\tilde{\mathcal{H}}$ denote the free algebra generated by these elements modulo the relations (A)–(C) in the proposition. We want to show that the natural homomorphism

$$\tilde{\mathcal{H}} \longrightarrow \mathcal{H}(G//B_k) \tag{1}$$

is an isomorphism.

Choose a set of reduced alternating words $w = s_{i_1}s_{i_2}\cdots s_{i_r}$ ($i_j \in \{1,2\}$, $i_j \neq i_{j+1}$) representing the elements of W_a, and define $\tilde{f}_w = \tilde{f}_{s_{i_1}} * \tilde{f}_{s_{i_2}} * \cdots * \tilde{f}_{s_{i_r}}$ (with $\tilde{f}_1$ for the trivial word). Let $\tilde{\mathcal{J}}$ denote the vector space span in $\tilde{\mathcal{H}}$ generated by the elements $\tilde{f}_g = \tilde{f}_x * \tilde{f}_w * \tilde{f}_y$, where $g = xwy$ for $x, y \in B$ and $w \in W_a$. As in [H], pages 38–39, the relations (B) and (C) imply that $\tilde{f}_g$ depends only on the double coset $B_k g B_k$, not on the particular representation $g = xwy$. Furthermore, $\tilde{\mathcal{J}}$ is invariant under left and right multiplication by any abstract generator $\tilde{f}_g$, $g \in S$: multiplication by $\tilde{f}_b$ ($b \in B$) is absorbed via B(ii), while multiplication by $\tilde{f}_{s_i}$ either increases the length of w or is reduced via C(i), C(ii), C(iii), and A(i)–(ii) exactly as in [H], pages 38–39.

From this invariance property and $\tilde{f}_1 \in \tilde{\mathcal{J}}$, it follows that $\tilde{\mathcal{J}} = \tilde{\mathcal{H}}$. Thus $\tilde{\mathcal{H}}$ is spanned by the elements $\tilde{f}_g$ indexed by the double cosets $B_k g B_k \in B_k\backslash G/B_k$. Since the images $f_g \in \mathcal{H}(G//B_k)$ of distinct double cosets have disjoint supports, they are linearly independent in $\mathcal{H}(G//B_k)$, which implies that the map (1) is injective.

Finally, since B_k is normal in B, for any $b_1, b_2 \in B$ and $w \in W_a$ we have

$$\mathrm{vol}(B_k b_1 w b_2 B_k) = \mathrm{vol}(b_1 B_k w B_k b_2) = \mathrm{vol}(B_k w B_k).$$

This and Lemma 4 imply $f_{b_1} * f_w * f_{b_2} = f_{b_1 w b_2}$. Combining this with the Iwahori-Bruhat decomposition $G = BW_aB$ and $f_w = f_{s_{i_1}} * \cdots * f_{s_{i_r}}$, we see that $\mathcal{H}(G//B_k)$ is spanned by the image of $\tilde{\mathcal{H}}$, so the map (1) is surjective. □

4. The structure of $\mathcal{H}(\overline{G}//B_k)_{gen}$

This section is the metaplectic analog of the previous section. We want to determine the structure of the metaplectic generalized Hecke algebra as a finitely generated algebra with generators and relations.

Throughout this section, we assume that $k > 0$ is chosen so large that $1 + \pi^k\mathcal{O} \subset (F^\times)^n$ (for instance, $k > 2\,\mathrm{ord}_\pi(n)$). As noted in Section 1, this guarantees that $(u, y) = 1$ for all $u \in 1 + \pi^k\mathcal{O}$ and all $y \in F^\times$, so that $\overline{B}_k$ splits. If the residual characteristic p of F is relatively prime to n, then $1 + \pi\mathcal{O} \subset (F^\times)^n$ already holds for $k = 1$, and in fact $\overline{B}$ splits.

Remark 2. Before stating Proposition 10, let us note how the n-fold cover behaves on the diagonal torus $T = \{h(a) \mid a \in F^\times\}$ and on B:

(1) For $h(a), h(b) \in T$, Kubota's cocycle gives $\beta(h(a), h(b)) = (b, a)$, so the commutator in $\overline{T}$ is $[(h(a), 1), (h(b), 1)] = (b, a)^2 = (b^2, a)$.
(2) When $n = 1$ or $n = 2$ (the classical 2-fold metaplectic cover), $(b, a)^2 = 1$ for all $a, b \in F^\times$, so $\overline{T}$ is abelian. Moreover, when $1 + \pi^k\mathcal{O} \subset (F^\times)^n$, for any $n \geq 1$ the subgroup $T_k = T \cap B_k = \{h(u) \mid u \in 1 + \pi^k\mathcal{O}\}$ satisfies $(u, a) = 1$ for all $a \in F^\times$, so $\overline{T}_k$ lies in the center of $\overline{T}$.
(3) For general $n \geq 1$, let $B^{(n)} = \{b \in B \mid$ the diagonal entries of b lie in $(\mathcal{O}^\times)^n(1 + \pi^k\mathcal{O})\}$. On $B^{(n)}$, all diagonal entries are n^{th} powers in $F^\times$, so the Hilbert symbol vanishes identically on $B^{(n)}$ and $\overline{B^{(n)}}$ splits and commutes with $(t_0, 1)$. In addition, choosing coset representatives of $B/B^{(n)} \cong \mathcal{O}^\times/(\mathcal{O}^\times)^n$ and of the double cosets $B_k\backslash G/B_k$, we may choose the coboundary $s : G \to \mu_n$ in Lemma 7 (with $s|_{B_k} \equiv 1$) so that β' trivializes the cocycle on the chosen coset decompositions.

Let the Haar measure on $\overline{G}$ be normalized so that $\overline{B}_k$ has measure one. Convolution gives $\mathcal{H}(\overline{G}//B_k)_{gen}$ the structure of an algebra.

Proposition 10. *Assume that* $\mathrm{vol}(\overline{B}_k) = 1$*, that* $k > 0$ *is chosen so large that* $1 + \pi^k \mathcal{O} \subset (F^\times)^n$*, and choose the equivalent cocycle* β' *via Lemma 7 as above. The algebra* $\mathcal{H}(\overline{G}//B_k)_{gen}$ *is generated by the functions* $\overline{f}_g$*,* $g \in S$*. On the* n^{th}*-power Iwahori subgroup* $B^{(n)}$ *(and on all of* B *when* $n \mid 2$ *and* $\gcd(p, n) = 1$*, or under the normalized coset splitting of Remark 2), these elements are subject to the same relations as in Proposition 8, except that* f_g *must be replaced by* $\overline{f}_g$*. Furthermore, these relations form a defining set of relations for* $\mathcal{H}(\overline{G}//B_k)_{gen}$ *as an algebra.*

Proof. First we verify that the claimed relations hold.

A(i), A(ii): By Lemma 6, we have

$$(\overline{f}_{s_i} * \overline{f}_{s_i})(x, 1) = \int_{B_k s_i B_k} \overline{f}_{s_i}(y^{-1}x, 1)\beta'(y, y^{-1}x)^{-1}\, dy.$$

Because $1 + \pi^k \mathcal{O} \subset (F^\times)^n$, β' is trivial on $U_k^- B_k$ and $U_{k-1}^+ B_k$, and for the same reason as in the non-metaplectic case, $\overline{f}_{s_i} * \overline{f}_{s_i}$ is supported on $\overline{((-1) \cdot (\mathrm{Ad}\, s_i)(B_k) \cdot B_k)}$. Evaluating at $(-I, 1)$ gives the same constant $c_i = \mathrm{vol}(B_k s_i B_k) = q$ as in the non-metaplectic case, which proves A(i) and A(ii).

B(i): For $g \in \overline{G}$ and $\phi \in \mathcal{H}(\overline{G}//B_k)_{gen}$, since $\phi(gx^{-1})\overline{f}_1(x) = \phi(g)$ for all $x = (b, \zeta) \in \overline{B}_k$ (by right B_k^*-invariance and genuineness of ϕ), we have

$$(\phi * \overline{f}_1)(g) = \int_{\overline{G}} \phi(gx^{-1})\overline{f}_1(x)\, dx = \int_{\overline{B}_k} \phi(gx^{-1})\overline{f}_1(x)\, dx = \phi(g)\,\mathrm{vol}(\overline{B}_k) = \phi(g).$$

Similarly, $\overline{f}_1 * \phi = \phi$, so $\overline{f}_1$ is the identity in $\mathcal{H}(\overline{G}//B_k)_{gen}$.

B(ii): By Lemma 6, we have

$$(\overline{f}_x * \overline{f}_y)(g, 1) = \int_{B_k x B_k} \overline{f}_y(h^{-1}g, 1)\beta'(h, h^{-1}g)^{-1}\, dh.$$

This is non-zero if and only if $gy^{-1} = xb$ for some $b \in B_k$, since B_k is normal in B. This implies $\mathrm{supp}(\overline{f}_x * \overline{f}_y) \subset \overline{B_k x y B_k}$, and Lemma 5 yields $\overline{f}_x * \overline{f}_y = \overline{f}_{xy}$.

B(iii): This is an immediate consequence of Lemma 5.

C(i): The proof of this identity is analogous to the proof in the non-metaplectic case (using Lemma 5 in place of Lemma 4 where appropriate) and is omitted.

C(ii), C(iii): These are proven in the same way as in the non-metaplectic case, so we only sketch the argument. The proof in the non-metaplectic case shows that each of these triple convolutions is supported on a single double $\overline{B}_k$-coset. On the other hand, the support of each non-zero function in the Hecke algebra contains at least one double $\overline{B}_k$-coset, since different double $\overline{B}_k$-cosets are disjoint. Since convolution preserves genuineness and bi-B_k^*-invariance, those functions whose support is precisely one double $\overline{B}_k$-coset are constant multiples of the corresponding $\overline{f}_g$. The constant multiple is evaluated at the double coset representative exactly as in the non-metaplectic case, and since the measures differ only by the normalized measure on μ_n, the volume computations are identical.

It remains to show that this list of relations is a defining set of relations for the algebra. Since $\{\overline{f}_g \mid B_k g B_k \in B_k \backslash G / B_k\}$ forms a vector space basis of $\mathcal{H}(\overline{G}//B_k)_{gen}$ indexed by the same double cosets as $\mathcal{H}(G//B_k)$, the argument is identical to the non-metaplectic case in Proposition 8. ∎

5. Application to representation theory

The following is our main result:

Theorem 11. *If $k > 0$ is chosen sufficiently large as in Proposition 10 (so in particular $\overline{B}_k$ splits), then the map $f_g \longmapsto \overline{f}_g$, $g \in S$, defines an isomorphism of algebras $\mathcal{H}(G//B_k) \to \mathcal{H}(\overline{G}//B_k)_{gen}$.*

Proof. This is an immediate consequence of Propositions 8 and 10. □

Corollary 12. *If $k > 0$ is as in Theorem 11, then the algebra isomorphism $\eta : \mathcal{H}(G//B_k) \to \mathcal{H}(\overline{G}//B_k)_{gen}$ induces a set-theoretic bijection $\eta^* : \mathcal{H}(\overline{G}//B_k)^\wedge_{gen} \to \mathcal{H}(G//B_k)^\wedge$ on the sets of equivalence classes of irreducible (genuine, in the case of $\overline{G}$) representations. Furthermore, if*

$$\begin{array}{ccc} \mathcal{H}(G//B_k) & \longrightarrow & \mathcal{H}(\overline{G}//B_k)_{gen} \\ f & \longmapsto & \overline{f} \end{array}$$

and

$$\begin{array}{ccc} \mathcal{H}(\overline{G}//B_k)^\wedge_{gen} & \longrightarrow & \mathcal{H}(G//B_k)^\wedge \\ \overline{\pi} & \longmapsto & \pi \end{array}$$

then $\mathrm{tr}(\overline{\pi}(\overline{f})) = \mathrm{tr}(\pi(f))$.

Proof. Let $\eta : \mathcal{H}(G//B_k) \to \mathcal{H}(\overline{G}//B_k)_{gen}$ be the algebra isomorphism of Theorem 11, and let $\eta^* : \mathcal{H}(\overline{G}//B_k)^\wedge_{gen} \to \mathcal{H}(G//B_k)^\wedge$ be the pullback map defined by $\eta^*(\overline{\pi}) = \overline{\pi} \circ \eta$. Suppose $\pi = \eta^*(\overline{\pi})$ and $\pi' = \eta^*(\overline{\pi}')$ are equivalent in $\mathcal{H}(G//B_k)^\wedge$. Since η is surjective, every $\overline{f} \in \mathcal{H}(\overline{G}//B_k)_{gen}$ is of the form $\overline{f} = \eta(f)$ for some $f \in \mathcal{H}(G//B_k)$, so any intertwining isomorphism between π and π' also intertwines $\overline{\pi}$ and $\overline{\pi}'$. Thus η^* is injective. Conversely, for any $(\pi, W) \in \mathcal{H}(G//B_k)^\wedge$, the composition $\overline{\pi} = \pi \circ \eta^{-1}$ defines an irreducible representation in $\mathcal{H}(\overline{G}//B_k)^\wedge_{gen}$ satisfying $\eta^*(\overline{\pi}) = \pi$, so η^* is surjective.

The equality of the traces $\mathrm{tr}(\overline{\pi}(\eta(f))) = \mathrm{tr}(\pi(f))$ holds by definition of η^*. □

Definition 13. Let K denote a compact open subgroup of $\overline{G}$. A locally constant complex-valued function f on $\overline{G}$ is called **left** (resp., **right**) **Hecke finite** if the vector space spanned by all functions of the form $h * f$ (resp., $f * h$), for $h \in \mathcal{H}(\overline{G}//K)$, is finite dimensional. Let $\mathcal{A}_{gen}(\overline{G})$ denote the space of all locally constant complex-valued genuine functions f on $\overline{G}$ for which

- f is double K-finite,
- f is right- and left-Hecke finite.

This space is called the space of **matrix coefficients** of $\overline{G}$. If (π, V) is a genuine admissible representation of $\overline{G}$ then the span of the elements $f(g) = \langle v_0^*, \pi(g)v_0 \rangle$ $(g \in \overline{G})$, for all $v_0 \in V$ and $v_0^* \in V^*$, the smooth contragredient of V, is called the space of **matrix coefficients of** π and is denoted $\mathcal{A}_{gen}(\pi)$.

Let ρ (resp., λ) denote the right (resp., left) regular representation of $\overline{G}$ on $\mathcal{A}_{gen}(\overline{G})$. The next proposition establishes that the metaplectic analogs of (the relevant parts of) Corollary 1.10.6 in [Sil] for $\overline{G}$ are true. The result below implies that we can "explicitly" realize the space of any genuine irreducible representation of $\overline{G}$ as a subspace of $\mathcal{A}_{gen}(\overline{G})$, a fact we will make use of later.

Proposition 14. *Let $f \in \mathcal{A}_{gen}(\overline{G})$ and let V_f denote the $\overline{G}$-module spanned by the translates $\rho(g)f$, $g \in \overline{G}$, and let π denote the restriction of ρ to V_f. We have:*

(a) *There exist elements $v_0 \in V_f$ and $v_0^* \in V_f^*$, the contragredient of V_f, such that $f(g) = \langle v_0^*, \pi(g)v_0 \rangle$, for all $g \in \overline{G}$.*
(b) *$\mathcal{A}_{gen}(\pi)$ is spanned by $\lambda(x)\rho(y)f$, $x, y \in \overline{G}$.*

Proof. The proof of this follows from the general arguments given in [Sil]. The results in that part of the book [Sil] hold for more general totally disconnected groups, so one need only check that the argument remains valid when "genuineness" is also assumed. □

We need to recall some facts regarding the relationship between smooth representations of $\overline{G}$ and finite dimensional $\mathcal{H}(\overline{G}//B_k)_{gen}$-modules. To this end, we recall briefly how one can construct smooth representations of $\overline{G}$ from finite dimensional $\mathcal{H}(\overline{G}//B_k)_{gen}$-modules—both the so-called "induced" and "produced" modules in §2 of Borel [Bo] will do this job. The "inverse" construction is then given along with some of the basic properties of these constructions in Proposition 16 below.

Definition 15. If (r, W) is a finite dimensional $\mathcal{H}(\overline{G}//B_k)$-module then we define $(I(r), I(W))$ to be the **induced** $\overline{G}$-module and $(P(r), P(W))$ to be the (smooth) **produced** $\overline{G}$-module constructed in [Bo]. Analogously, if (r, W) is a finite dimensional $\mathcal{H}(\overline{G}//B_k)_{gen}$-module then we define $(I_{gen}(r), I_{gen}(W))$ to be the **induced** genuine $\overline{G}$-module and $(P_{gen}(r), P_{gen}(W))$ to be the **produced** genuine $\overline{G}$-module obtained by replacing $C_c(\overline{G}/B_k)$ by $C_c(\overline{G}/B_k)_{gen}$ in the constructions in [Bo].

Proposition 16. *Let (r, W) be a finite dimensional $\mathcal{H}(\overline{G}//B_k)_{gen}$-module and let (π, V) be a genuine smooth representation of $\overline{G}$.*

(a) *There are natural $\mathcal{H}(\overline{G}//B_k)_{gen}$-module isomorphisms*
$$W \cong I_{gen}(W)^{B_k} \quad \text{and} \quad W \cong P_{gen}(W)^{B_k}.$$
(b) *$I_{gen}(W)$ is generated, as a $\overline{G}$-module, by $I_{gen}(W)^{B_k}$.*
(c) *The natural restriction maps yield isomorphisms*
$$\rho_P : \mathrm{Hom}_{\overline{G}}(V, P_{gen}(W)) \longrightarrow \mathrm{Hom}_{\mathcal{H}(\overline{G}//B_k)_{gen}}(V^{B_k}, W)$$
and
$$\rho_I : \mathrm{Hom}_{\overline{G}}(I_{gen}(W), V) \longrightarrow \mathrm{Hom}_{\mathcal{H}(\overline{G}//B_k)_{gen}}(W, V^{B_k}).$$
(d) *(r, W) is an irreducible $\mathcal{H}(\overline{G}//B_k)_{gen}$-module if and only if $(I_{gen}(r), I_{gen}(W))$ is an irreducible representation of $\overline{G}$. Moreover, if (r, W) is an irreducible $\mathcal{H}(\overline{G}//B_k)_{gen}$-module then $I_{gen}(r) \cong P_{gen}(r)$.*

Proof. In the case $n = 1$, the statements in the proposition above are either in §§1–2 of [Bo] as stated or follow from Borel's results using Schur's lemma (see, for example, Lemma 3.14 in [CR]). In general, one need only check that the argument remains valid when "genuineness" is also assumed. □

Definition 17. We define a genuine admissible irreducible representation of $\overline{G}$ to be **supercuspidal** if and only if its matrix coefficients are compactly supported. We define a genuine admissible irreducible representation of $\overline{G}$ to be **square-integrable** if and only if its matrix coefficients belong to $L^2(\overline{G})$.

We define an irreducible finite dimensional $\mathcal{H}(\overline{G}//B_k)_{gen}$-module (r,W) to be **supercuspidal** (resp., **square-integrable**) if and only if $I_{gen}(r)$ is a supercuspidal (resp., square-integrable) representation of $\overline{G}$.

Next we observe that, as in the case of $GL(n)$, the correspondence of Corollary 12 preserves supercuspidals and discrete series representations ([H], page 33).

Corollary 18. *Assume $k>0$ is as in Corollary 12.*

- *An admissible genuine representation $\overline{\pi}\in\mathcal{H}(\overline{G}//B_k)^{\wedge}_{gen}$ is supercuspidal if and only if π is, where $\pi,\overline{\pi}$ are as in Corollary 12.*
- *$\overline{\pi}$ is square-integrable if and only if π is, where $\pi,\overline{\pi}$ are as in Corollary 12.*

Proof. Suppose that the mapping $\mathcal{H}(\overline{G}//B_k)^{\wedge}_{gen}\to\mathcal{H}(G//B_k)^{\wedge}$ sends a representation $(\overline{r},\overline{W})$ to a representation (r,W). Note that the construction of the correspondence in Corollary 12 implies that $W=\overline{W}$. From the construction of $I_{gen}(\overline{W})=C_c(\overline{G}/B_k)_{gen}\otimes_{\mathcal{H}(\overline{G}//B_k)_{gen}}\overline{W}$ (more precisely, from the definition of $\otimes_{\mathcal{H}(\overline{G}//B_k)_{gen}}$), we see that the $\overline{r}(\overline{f}_g)$-action on $\overline{W}\cong I_{gen}(\overline{W})^{B_k}$ corresponds to the $\overline{f}_g$-action (via convolution) on $C_c(\overline{G}/B_k)_{gen}$. Identifying $\overline{W}$ with $I_{gen}(\overline{W})^{B_k}$, we see that each element of $\overline{W}$ is Hecke-finite and may therefore be regarded as a matrix coefficient of $\overline{G}$. Let $(\overline{\pi},\overline{V})$ denote the $\overline{G}$-module generated by an element $w\in\overline{W}\cong I_{gen}(\overline{W})^{B_k}$. By Proposition 14 (or Proposition 16(b)), $\overline{\pi}=I_{gen}(\overline{r})$.

From this and the fact that $W=\overline{W}$, it follows that if the elements of $\overline{W}$ are compactly supported (when regarded as matrix coefficients of $\overline{G}$) then the elements of W are also compactly supported (when regarded as matrix coefficients of G). Now assume that $(\overline{r},\overline{W})$ is supercuspidal, so the matrix coefficients of $I_{gen}(\overline{r})$ have compact support. As indicated above, (r,W) is supercuspidal as well, and the converse holds symmetrically. This establishes the first part of the corollary.

The second part of the corollary is proved similarly, hence is omitted. □

6. The Iwahori algebra

In the case where $k=0$ and $\gcd(p,2n)=1$, the genuine Iwahori-Hecke algebra $\mathcal{H}(\overline{G}//B)_{gen}$ is described by a general theorem due to Savin [S]. Recall that for $G=SL(2,F)$, the dual group of the n-fold metaplectic cover $\overline{G}$ depends on the parity of n: let $G^\vee=SL(2,F)$ when n is odd, and let $G^\vee=PGL(2,F)\cong SO(3,F)$ when n is even (with Iwahori subgroup $B^\vee\subset G^\vee$). We recall Savin's theorem in this setting:

Theorem 19 (Savin [S])**.** *Assume* $\gcd(p,2n)=1$. *There is an isomorphism of Iwahori-Hecke algebras*

$$\mathcal{H}(\overline{G}//B)_{gen}\xrightarrow{\sim}\mathcal{H}(G^\vee//B^\vee),$$

where $G^\vee=G=SL(2,F)$ when n is odd, and $G^\vee=PGL(2,F)$ when n is even.

Corollary 20. *Assume* $\gcd(p,2n)=1$. *The isomorphism $\mathcal{H}(\overline{G}//B)_{gen}\to\mathcal{H}(G^\vee//B^\vee)$ induces a set-theoretic bijection $\mathcal{H}(\overline{G}//B)^{\wedge}_{gen}\to\mathcal{H}(G^\vee//B^\vee)^{\wedge}$ on the sets of equivalence classes of irreducible (genuine, in the case of $\overline{G}$) representations with a B-fixed (resp., $B^\vee$-fixed) vector. Furthermore, if*

$$\begin{array}{ccc}\mathcal{H}(G^\vee//B^\vee) & \longrightarrow & \mathcal{H}(\overline{G}//B)_{gen}\\ f & \longmapsto & \overline{f}\end{array}$$

and

$$\begin{array}{ccc} \mathcal{H}(\overline{G}//B)^{\wedge}_{gen} & \longrightarrow & \mathcal{H}(G^{\vee}//B^{\vee})^{\wedge} \\ \overline{\pi} & \longmapsto & \pi \end{array}$$

then $\mathrm{tr}(\overline{\pi}(\overline{f})) = \mathrm{tr}(\pi(f))$.

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